

ROSHNI BAGGA

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EDUCATION

Thapar Institute of Engineering & Technology

B.Tech, Computer Engineering

Aug 2024 – May 2028

GPA: 9.43 / 10.0

Coursework: Data Structures & Algorithms | Machine Learning | DBMS | Operating Systems | OOP

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, HTML/CSS
Frameworks	FastAPI, React, Flask, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, BERT
Tools & Cloud	Git, GitHub, Vercel, REST APIs, Jupyter, Claude API, VS Code, Agile/Scrum
Domains	Data Products Data Governance Data Quality Product Development NLP Financial Analytics

PROJECTS

RepoLens

Jan 2026 – Present

Python | FastAPI | Claude API | GitHub API | React | Vercel | [Live App](#) | [GitHub](#)

- Owned full product development lifecycle – ideation to product launch – of a live AI data product that ingests any GitHub repository and surfaces **3+ analytical insight categories** (code quality, contributor patterns, activity trends) for real users.
- Architected a scalable REST API backend (FastAPI) with GitHub API integration, deployed on Vercel with **100% uptime**; designed for global configurability and reuse, enabling extension to enterprise-scale data products.
- Applied data governance principles to output design – ensuring reproducible, auditable natural-language insights via Claude API across **multiple repository types**.

Cache – Credit Card Fraud Detection

Aug 2025 – Oct 2025

Python | XGBoost | Scikit-learn | Pandas | SMOTE | Matplotlib | [GitHub](#)

- Solved a complex business problem in financial risk: benchmarked **4** algorithms (XGBoost, LR, RF, SVM) on a **284K**-transaction dataset, achieving **86.76% recall** – directly applicable to business decisions affecting millions of cardmember accounts.
- Implemented rigorous data quality controls and applied SMOTE oversampling to resolve severe class imbalance (**0.17%** fraud rate), improving minority-class detection by **34%** across highly skewed real-world financial datasets.
- Delivered a modular, API-ready data product with configurable architecture supporting **4 interchangeable ML models**, built for scalable product launch and reuse across enterprise financial platforms.

HireIQ – AI Resume Analyzer

Nov 2025 – Dec 2025

Python | BERT | NLP | Flask | Scikit-learn | PyPDF2 | [GitHub](#)

- Designed and executed product launch of an NLP-powered data product performing ATS keyword-gap analysis across **10+ resume-JD pairs**, generating structured optimization reports end-to-end.
- Integrated BERT-based semantic similarity scoring across **2 analysis dimensions** (keyword match + contextual relevance), achieving measurably higher accuracy vs. keyword-only tools.
- Enforced data governance and data quality standards throughout – clean pipelines, reproducible outputs, compliance-ready design – aligned with managing data assets used in business decisions at scale.

EXPERIENCE

Open Source Contributor – GirlScript Summer of Code (GSSoC)

May 2025 – Aug 2025

- Shipped **5+** production-merged contributions to open-source ML repositories across **3+** projects – fixing critical inference bugs and improving pipeline throughput in a distributed Agile, cross-functional team environment.
- Led cross-functional code reviews with **10+** global contributors, establishing data quality gates and mentoring peers – building stakeholder management and team coordination skills central to product development roles.
- Authored technical documentation adopted as standards across **2** repositories, improving new contributor onboarding and demonstrating structured communication and process ownership.

CERTIFICATIONS & ACHIEVEMENTS

AWS Solutions Architecture Job Simulation – Forage

2025

Deloitte Data Analytics Job Simulation – Forage

2025